

Sabhyata Sinha

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Projects

AG-USegNet: Attention-Guided Ultrasound Segmentation — Deep Learning, C++ Deployment [GitHub](#)

- Designed a hybrid **U-Net + SegNet architecture** with attention-gated skip connections for low-SNR ultrasound segmentation.
- Improved segmentation performance to **F1: 0.8856, IoU: 0.7947**, achieving **+25% gain** over baseline via ablation across 5 architectures.
- Built a **cross-language inference pipeline (Python → ONNX → C++)**, enabling portable deployment across Windows and Linux systems.
- Engineered an end-to-end pipeline (preprocessing, inference, post-processing) optimized for **near real-time inference**.

PharmaSage AI – Drug Repurposing Platform — Python, Streamlit [GitHub](#)

- Developed an **agentic AI system** integrating clinical, research, and patent datasets for drug repurposing insights.
- Built an interactive **Streamlit dashboard** with automated reporting and data visualization.
- Designed data pipelines to enable **multi-source aggregation and decision support**.

Online Payment Fraud Detection — Machine Learning [GitHub](#)

- Trained and evaluated **XGBoost, LSTM, and Random Forest models** on imbalanced transaction datasets.
- Improved detection performance using **feature engineering and resampling techniques**.
- Optimized **precision-recall tradeoff** to reduce false positives in high-risk transactions.

Skills

Languages: Python, C/C++, SQL

ML & AI: Deep Learning, CNNs, Computer Vision, LLMs, Model Evaluation, Feature Engineering

Frameworks/Libraries: TensorFlow, Keras, scikit-learn, NumPy, pandas, OpenCV, ONNX

Systems & Tools: FastAPI, Streamlit, Git, Linux, Jupyter

Core Concepts: Image Segmentation, Model Deployment, SDLC, Debugging

Systems: Multithreading, Memory Management

Experience

Tiaro Rides (AIC-MUJ Hybrid) — *AI/ML Intern*

May 2025 – Nov 2025

- Reduced fare prediction error by **40% (RMSE)** using feature engineering on vehicle and usage attributes.
- Deployed real-time inference services via **FastAPI REST APIs** and interactive dashboards using Streamlit.
- Integrated **C++ inference modules** with Python pipelines for low-latency predictions.
- Built a **face recognition system** using OpenCV for driver verification and fraud prevention.

Plasmid Innovation Pvt. Ltd. — *Machine Learning Intern*

June 2024 – Aug 2024

- Built end-to-end ML pipelines including **data preprocessing, feature scaling, and model evaluation**.
- Contributed to a **fraud detection system** for online transactions.

Bharat Sanchar Nigam Limited (BSNL) — *Vocational Trainee (Telecom & Networking)*

Dec 2025 – Jan 2026

- Studied **telecom infrastructure** including switching systems, FTTH, and OFC networks.
- Analyzed **routing mechanisms and exchange operations** in real-world communication systems.
- Observed **network maintenance and fault diagnosis workflows**.

Education

Manipal University Jaipur — B.Tech in CSE (AI/ML)

Aug 2023 – July 2027

CGPA: 7.8

Achievements

- Contributor in **SWOC**; submitted pull requests and resolved issues in open-source projects.
- Open Source Contributor at **PWOC**; developed a **DenseNet + NASNet hybrid model**.
- Advanced to **Round 2, EY Techathon 6.0**.
- Selected for **McKinsey Forward Program**.
- Built **LearnEx**, an AI-based skill exchange platform (Ideathon project).